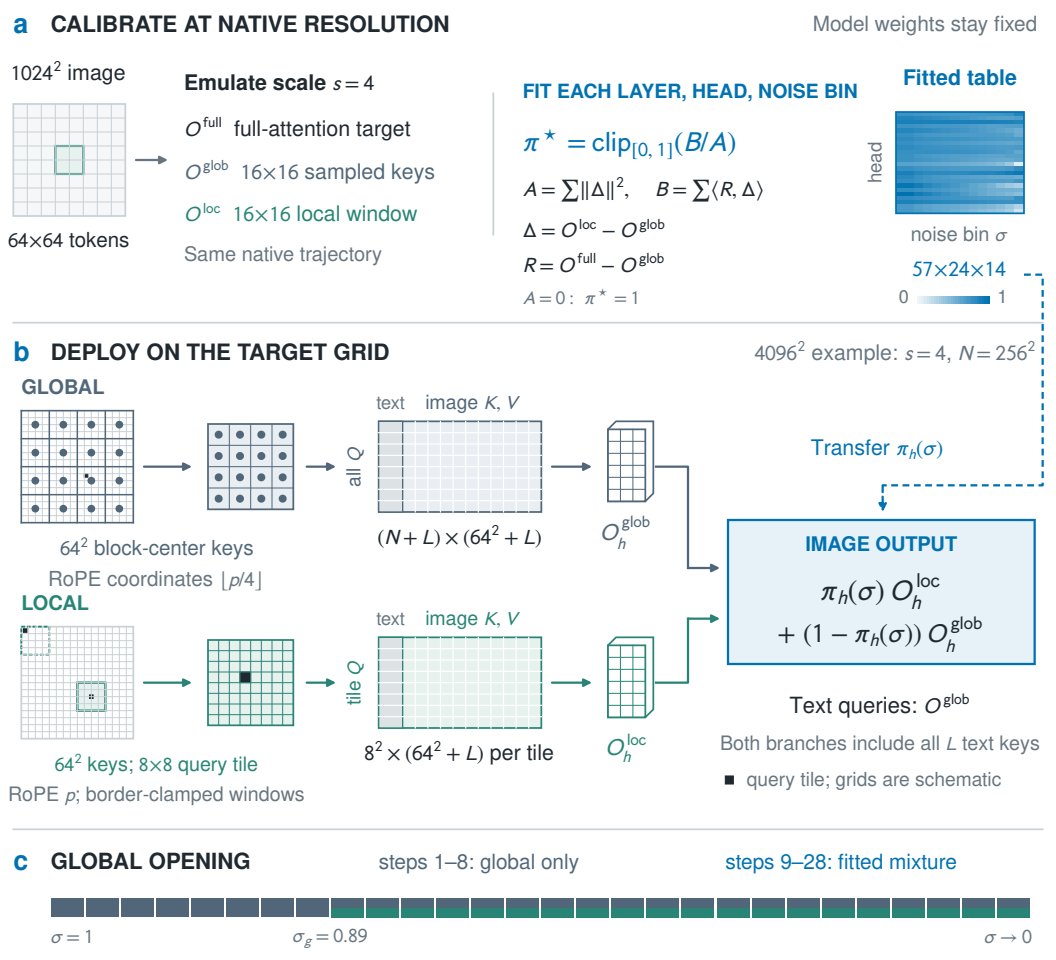
a CALIBRATE AT NATIVE RESOLUTION
Model weights stay fixed
1024² image
64×64 tokens
Emulate scale s = 4
O^full full-attention target
O^glob 16×16 sampled keys
O^loc 16×16 local window
Same native trajectory
FIT EACH LAYER, HEAD, NOISE BIN
π* = clip[0, 1](B/A)
A = Σ‖Δ‖², B = Σ⟨R, Δ⟩
Δ = O^loc − O^glob
R = O^full − O^glob
A = 0 : π* = 1
Fitted table
head
noise bin σ
57×24×14
0
1
b DEPLOY ON THE TARGET GRID
4096² example: s = 4, N = 256²
GLOBAL
64² block-center keys
RoPE coordinates ⌊p/4⌋
text
image K, V
all Q
(N + L) × (64² + L)
O_h^glob
Transfer π_h(σ)
IMAGE OUTPUT
π_h(σ) O_h^loc
+ (1 − π_h(σ)) O_h^glob
LOCAL
64² keys; 8×8 query tile
RoPE p; border-clamped windows
text
image K, V
tile Q
8² × (64² + L) per tile
O_h^loc
Text queries: O^glob
Both branches include all L text keys
query tile; grids are schematic
c GLOBAL OPENING
steps 1–8: global only
steps 9–28: fitted mixture
σ = 1
σ_g = 0.89
σ → 0